Polycystic ovary syndrome (PCOS)

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Microbiome Signatures, Microbiome Medicine

Abstract. Polycystic ovary syndrome (PCOS) is a common and complex endocrine and metabolic disorder affecting women of reproductive age.

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Overview

Polycystic ovary syndrome (PCOS) is a common and complex endocrine and metabolic disorder affecting women of reproductive age, with a global prevalence ranging from 4% to 20% depending on the diagnostic criteria used.[1][2] It is typically characterized by the presence of hyperandrogenism, ovulatory dysfunction, and polycystic ovarian morphology,[3] and is strongly associated with various health issues including infertility, insulin resistance (IR), type 2 diabetes mellitus, and cardiovascular disease.[4][5] Although the precise pathogenesis of PCOS is not fully understood, accumulating evidence suggests a complex interplay of genetic predisposition, epigenetic modifications, and environmental factors such as dietary choices and lifestyle,[6] with emerging research highlighting the causal role of gut dysbiosis in its development. [7][8][9]

Diagnosis

PCOS diagnosis typically involves the identification of a combination of clinical and biochemical features, primarily focusing on hyperandrogenism, ovulatory dysfunction including irregular menstrual cycles, infrequent ovulation, or anovulation, and polycystic ovarian morphology (PCOM) on ultrasound.[10] The Rotterdam criteria requires at least two out of these three features for diagnosis, after excluding other causes, as ovarian cysts alone are not enough for a diagnosis, as they can occur in other conditions. [11]

Causal Theories

The pathogenesis of PCOS is multifactorial, involving genetic, environmental, and hormonal factors. Insulin resistance is a central feature, and hyperinsulinemia leads to increased ovarian androgen production.[12] This can disrupt normal follicular development and contribute to the clinical manifestations of PCOS. Environmental factors, including diet and lifestyle, can exacerbate insulin resistance, while genetic predispositions to hyperandrogenism and metabolic dysfunction further complicate the condition. [13] There is emerging evidence supporting the role of gut microbiota in the development and progression of PCOS, with altered microbial compositions being linked to insulin resistance and inflammation.[14]

Associated Conditions

PCOS is a complex condition with significant associations to a variety of metabolic, hormonal, and psychological comorbidities. These include infertility, which is a hallmark feature due to anovulation, metabolic syndrome, and insulin resistance, which elevate the risk of type 2 diabetes.[15][16] Women with PCOS also face a higher risk of cardiovascular diseases, such as hypertension and dyslipidemia, due to hormonal imbalances and metabolic dysfunctions.[17] Endometrial cancer is another concern, as prolonged anovulation leads to unopposed estrogen exposure, increasing the risk of endometrial hyperplasia.[18] Additionally, women with PCOS are more likely to experience depression, anxiety, and eating disorders, potentially exacerbated by physical symptoms like hirsutism, acne, and weight gain.

Primer

To understand PCOS comprehensively, it is essential to explore the intersections of hormonal dysregulation, insulin resistance, and the gut microbiome. The elevated levels of insulin in PCOS not only contribute to metabolic abnormalities but also directly affect ovarian steroidogenesis, exacerbating the symptoms of hyperandrogenism.[19] Dysbiosis in the gut microbiome is now being recognized as a significant contributor to the inflammation and metabolic dysfunction observed in PCOS. Studies suggest that gut microbial imbalances may lead to the development of insulin resistance through mechanisms such as the leakage of lipopolysaccharide (LPS) into the bloodstream, which in turn activate immune pathways that interfere with insulin receptor function.[20][21] This relationship between the gut microbiome and insulin resistance is crucial to understanding the broader pathophysiology of PCOS, particularly regarding the gut-brain axis and hormonal regulation and presents opportunities for microbiome-based diagnostics and microbiome-targeted interventions (MBTIs).[22]

Metallomic Signature

Heavy metals and trace elements play a significant role in the pathophysiology of polycystic ovary syndrome (PCOS). Studies have revealed imbalances in several essential trace elements such as zinc, copper, manganese, and the accumulation of toxic metals like lead (Pb), and cadmium (Cd) in PCOS patients. These elements contribute to oxidative stress (OS), inflammation, and hormonal dysregulation, which are hallmarks of PCOS. Elevated zinc and nickel levels, in particular, are indicative of inflammation and metabolic dysfunction, which are key components of PCOS pathology. By examining the metallomic signature alongside microbial shifts in PCOS, it becomes clear that targeting these imbalances could offer novel therapeutic avenues, including dietary interventions and chelation therapies, to manage oxidative stress and inflammation in PCOS patients.

▶ **What are the metallomic signatures of PCOS?**

Nutritional Immunity

In PCOS, imbalances in essential micronutrients like zinc, copper, and manganese contribute to immune dysfunction and inflammation. Deficiencies in these nutrients are linked to increased oxidative stress and metabolic issues, such as insulin resistance and hormonal disturbances.[44][45][x][46] In addition to these micronutrients, calprotectin, a protein involved in inflammation and immune response, plays a critical role in modulating the immune system in PCOS. Elevated calprotectin levels have been associated with increased inflammation and oxidative stress in PCOS patients, further highlighting the importance of balancing trace elements and inflammatory markers.[47] Addressing these nutritional imbalances and the

inflammatory pathways, including calprotectin regulation, through targeted interventions could help manage inflammation, improve metabolic function, and restore microbial balance, offering potential benefits for PCOS management.

Interventions

Our validation method confirms the effectiveness of microbiome-targeted interventions (MBTIs) by linking their therapeutic impact on microbial imbalances with clinical outcomes, key pathological markers, and major microbial associations (MMAs) associated with PCOS. This dual approach ensures that interventions effectively target both the microbial imbalances and the fundamental biological processes of PCOS, reinforcing the precision of microbiome signatures and the success of MBTIs. Moreover, promising candidates demonstrating preliminary benefits or partial alignment are highlighted for further exploration, contributing to a pipeline of novel treatments for future validation.

Intervention	Classification	MBTI Status
Fecal Microbiota Transplantation (FMT)	Microbiota-based therapy	Validated
Metformin	Drug Repurposing	Validated
Curcumin	Supplement	Validated
Resistant Dextrin	Supplement	Validated
Statin Therapy	Pharmaceutical	Validated
Probiotics	Supplement	Validated
Synbiotics	Supplement	Validated

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